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13	Implement the Car Price Prediction Model using Python

CSV FILE

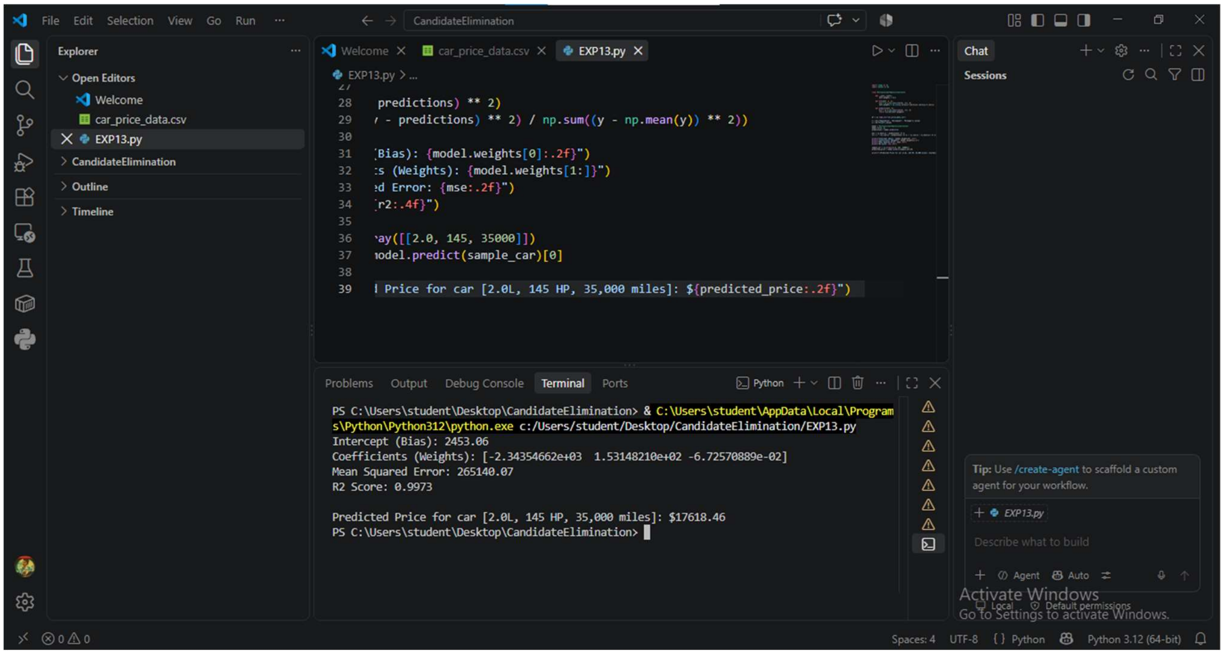
```
EngineSize,Horsepower,Mileage,Price
1.6,105,45000,12000
2.0,150,30000,18500
2.5,190,20000,25000
1.4,90,60000,9500
3.0,250,15000,32000
2.0,140,40000,16000
1.8,125,50000,13500
3.5,300,10000,40000
```

PROGRAM

```python
import numpy as np
import pandas as pd



class MultipleLinearRegressionScratch:

    def __init__(self):
        self.weights = None

    def fit(self, X, y):
        X_b = np.c_[np.ones((len(X), 1)), X]
        self.weights = np.linalg.inv(X_b.T.dot(X_b)).dot(X_b.T).dot(y)
```

```python
    def predict(self, X):
        X_b = np.c_[np.ones((len(X), 1)), X]
        return X_b.dot(self.weights)



df = pd.read_csv("car_price_data.csv")

X = df[["EngineSize", "Horsepower", "Mileage"]].values
y = df["Price"].values

model = MultipleLinearRegressionScratch()
model.fit(X, y)
predictions = model.predict(X)

mse = np.mean((y - predictions) ** 2)
r2 = 1 - (np.sum((y - predictions) ** 2) / np.sum((y - np.mean(y)) ** 2))

print(f"Intercept (Bias): {model.weights[0]:.2f}")
print(f"Coefficients (Weights): {model.weights[1:]}")
print(f"Mean Squared Error: {mse:.2f}")
print(f"R2 Score: {r2:.4f}")

sample_car = np.array([[2.0, 145, 35000]])
predicted_price = model.predict(sample_car)[0]

print(f"\nPredicted Price for car [2.0L, 145 HP, 35,000 miles]: ${predicted_price:.2f}")
```

OUTPUT


PS C:\Users\student\Desktop\CandidateElimination> & C:\Users\student\AppData\Local\Programs\Python\Python312\python.exe c:/Users/student/Desktop/CandidateElimination/EXP13.py
Intercept (Bias): 2453.06
Coefficients (Weights): [-2.34354662e+03 1.53148210e+02 -6.72570889e-02]
Mean Squared Error: 265140.07
R2 Score: 0.9973
Predicted Price for car [2.0L, 145 HP, 35,000 miles]: $17618.46
PS C:\Users\student\Desktop\CandidateElimination>